No.	Status	Date
CIE-FFS-2025-004-R0	Issued	2025-02-17

Fitness-for-Service Assessment - 4-RA-101 Local Metal Loss (Level 2)

Scope and Responsibilities

This document provides a Level 2 Fitness-for-Service assessment for line 4-RA-101 within the ATU-100 at the Hamilton Plant. The assessment addresses localized metal loss identified at 4-RA-101-06, where wall thickness has degraded below the design minimum. The evaluation follows the methodologies prescribed in API 579-1/ASME FFS-1 to determine the structural integrity of the piping system under design conditions of 862 kPa(g) and 82 °C.

Priya Natarajan, acting as the FFS engineer for Carrow Integrity Engineering Ltd., is responsible for the engineering analysis and the determination of the remaining strength factor. The Hamilton Plant inspection department is responsible for the implementation of the recommended monitoring intervals and the execution of the final repair scope during the scheduled turnaround.

Background and Inspection Results

During the five-year survey of the ATU-100, thickness measurements recorded in LNDE-RPT-20241025-031-R2 identified significant thinning at 4-RA-101-06. The reading of 3.59 mm was found to be below the t-min of 3.81 mm, triggering QSCP-NCR-2024-031. Subsequent high-resolution PAUT corrosion mapping documented in LNDE-RPT-20241202-036-R0 confirmed a minimum thickness of 3.55 mm, indicating an accelerated corrosion rate compared to historical data.

Historical records show a stable corrosion rate of 0.029 mm/y until 2016. Following a 2019 trim change to the upstream let-down valve, which was performed under a work order without a formal Management of Change (MOC), the rate increased to 0.253 mm/y and recently peaked at 0.659 mm/y. This localized attack is attributed to flashing downstream of the modified valve trim, which has significantly reduced the remaining life of the A106 Gr.B material.

Fitness-for-Service Evaluation

The Level 2 assessment per API 579-1/ASME FFS-1 was conducted using the PAUT data from LNDE-RPT-20241202-036-R0. Although the current thickness of 3.55 mm is below the t-min required by ASME B31.3 for new construction, the calculated Remaining Strength Factor (RSF) exceeds the allowable limit for continued operation at the design pressure of 862 kPa(g). The localized nature of the flashing-induced erosion-corrosion allows for temporary service provided that the degradation mechanism is monitored.

The analysis confirms that the piping remains acceptable for service at design temperature and pressure. However, the current short-term corrosion rate of 0.659 mm/y necessitates a reduction in the inspection interval to ensure the wall thickness does not breach the structural stability limits before the next major maintenance opportunity.

Recommendations and Conclusion

It is recommended to restore the original let-down valve trim to mitigate the flashing conditions causing the accelerated metal loss at 4-RA-101-06. The line is considered fit for continued service at design conditions until the August 2026 turnaround. A mandatory re-measurement of the affected area must be performed within 6 months to validate that the corrosion rate has not further accelerated.

Final remediation requires the replacement of the affected elbow and downstream piping section during the August 2026 turnaround. All replacement activities must comply with ASME B31.3. Until replacement, the Hamilton Plant must maintain the current operating envelope and ensure no increases in pressure or temperature occur that exceed the design limits evaluated in this CIE-FFS-2025-004-R0 report.

Sign-off	Name	Function	Date	Signature
Prepared	Priya Natarajan	FFS engineer, P.Eng.	2025-02-17	_____